



UNIVERSITY
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Efficient Data Storage Using Programmable Containers in Edge Computing

Pedro Ojo Oriakhi

Master's in Telecommunications and Computer Engineering

Supervisors:

Ph.D. Professor José Moura, Assistant Professor,
Iscte - Instituto Universitário de Lisboa

Ph.D. Professor Pedro Ramos, Assistant Professor,
Iscte - Instituto Universitário de Lisboa

Month, Year

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TECHNOLOGY
AND ARCHITECTURE

Department of Information Science and Technology

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Acknowledgments

Write here your acknowledgements and, if applicable, any grants or sources of funding.

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Resumo

Palavras-Chave:

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Abstract

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Keywords: *Keywords; separated; by; semicolons*

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List of Acronyms

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1.1. Context and Motivation

1.2. Objectives

1.3. Research Questions

1.4. Methodology Overview

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Literature Review

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2.2. Edge Computing Architectures

2.3. SDN and Programmable Infrastructure

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2.5. Metadata-Driven Data Placement and Replication

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CHAPTER 3

System Architecture

3.1. Design Principles

3.2. Architecture Overview

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- 5.2. Edge Service and Storage
- 5.3. Network Infrastructure
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CHAPTER 6

Results and Discussion

- 6.1. RQ1 — Telemetry Freshness and Delivery Cadence
- 6.2. RQ2 — Metadata-Aware Backend Selection
- 6.3. RQ3 — Data Locality Strategies
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CHAPTER 7

Conclusions and Future Work

7.1. Summary of Findings

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